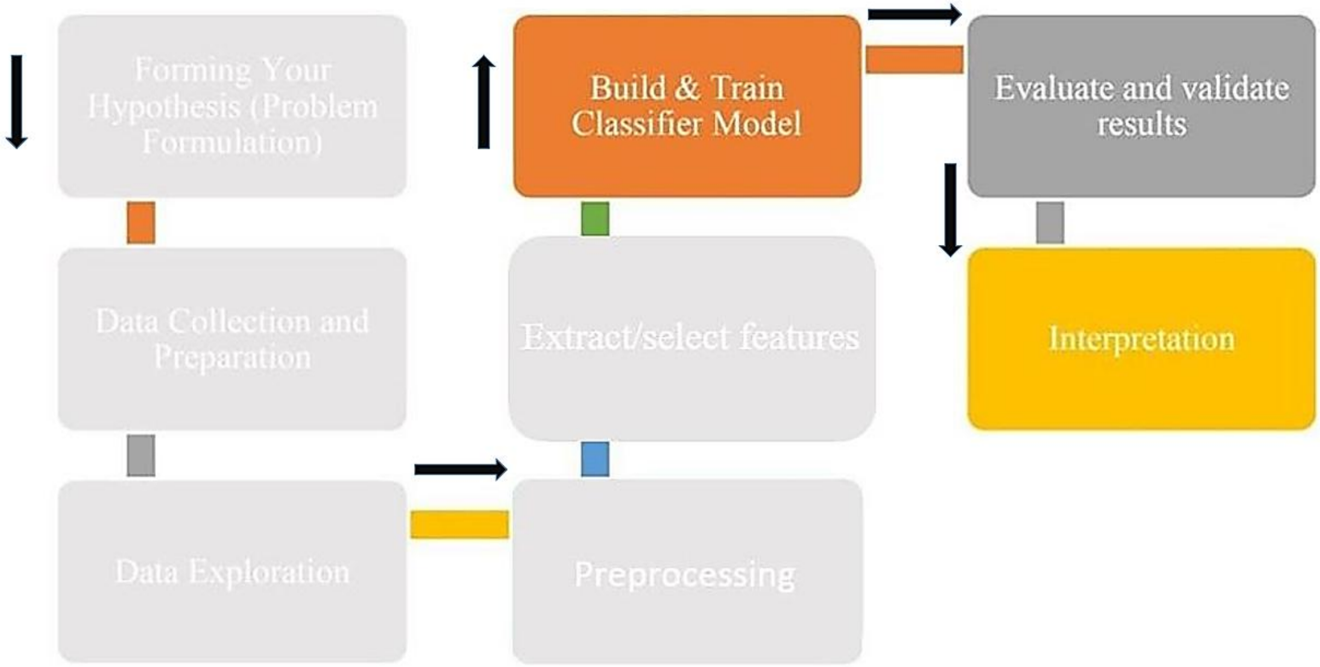
ELSEWEDY UNIVERSITY OF TECHNOLOGY
- POLYTECHNIC OF EGYPT -

Algorithms for Data Science

Lecture 7: Building ML model (Part II)

© Spring 2026 – **Dr. Dalia Ezzat**
dalia.ezzat@sut.edu.eg

The Data Science Pipeline



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Naïve Bayes

- Naïve Bayes is a probabilistic classifier based on Bayes' Theorem.

Key Formula (Bayes' Theorem):

$$P(Y|X) = \frac{P(X|Y) \cdot P(Y)}{P(X)}$$

- $P(Y|X)$: Probability of class Y given features X (**Posterior**).
- $P(X|Y)$: Probability of features X given class Y (**Likelihood**).
- $P(Y)$: Prior probability of class Y.
- $P(X)$: Probability of features X (normalization constant).

How Naïve Bayes Works?

Step 1: Calculate Prior Probability ($P(Y)$)

- Count how often each class appears in the training data.
- Example:
  - In a spam detection dataset with 60% spam and 40% non-spam:

$$P(\text{Spam}) = 0.6, \quad P(\text{Not Spam}) = 0.4$$

Step 2: Calculate Likelihood ($P(X|Y)$)

- For each feature, compute its probability given the class.
- Example (for the word "FREE" in spam emails):
  - If "FREE" appears in 30 out of 100 spam emails:

$$P(\text{"FREE"}|\text{Spam}) = \frac{30}{100} = 0.3$$

How Naïve Bayes Works?

Step 3: Apply Naïve Assumption

- Assume features are **independent**, so:

$$P(X_1, X_2, \ldots, X_n|Y) = P(X_1|Y) \cdot P(X_2|Y) \cdots P(X_n|Y)$$

Step 4: Compute Posterior Probability ($P(Y|X)$)

- For a new email with words $X = \{"FREE", "Win"\}$:

$$P(\text{Spam}|X) \propto P(\text{"FREE"}|\text{Spam}) \cdot P(\text{"Win"}|\text{Spam}) \cdot P(\text{Spam})$$

Step 5: Predict the Class

- Choose the class with the **highest posterior probability**.

Types of Naïve Bayes Classifiers

- **Naïve Bayes algorithms are categorized based on how they handle different data types. Here are the main types with examples:**

1. Gaussian Naïve Bayes

- **Use Case**: Continuous numerical data (assumes features follow a normal distribution).
- **Example**:
  - *Diabetes prediction* (Glucose levels, BMI, Age).
  - *Formula*:

$$P(x_i|Y) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} e^{-\frac{(x_i-\mu_Y)^2}{2\sigma_Y^2}}$$

where μ_Y = mean, σ_Y = standard deviation for class Y.

Types of Naïve Bayes Classifiers

- **Naïve Bayes algorithms are categorized based on how they handle different data types. Here are the main types with examples:**

2. Multinomial Naïve Bayes

- **Use Case**: Discrete counts (e.g., word frequencies in text).
- **Example**:
  - Spam detection (counts of words like "FREE", "Win").
- **Formula**:

$$P(x_i|Y) = \frac{\text{Count}(x_i, Y) + \alpha}{\text{Total count in } Y + \alpha k}$$

  - $\text{Count}(x_i, Y)$: Number of times feature x_i appears in class Y
  - $\text{Total count in } Y$: Sum of all feature counts in class Y
  - k: Number of unique features (vocabulary size)
  - α: Smoothing parameter ($\alpha = 1$ for Laplace smoothing)

Types of Naïve Bayes Classifiers

- **Naïve Bayes algorithms are categorized based on how they handle different data types. Here are the main types with examples:**

4. Categorical Naïve Bayes

- **Use Case**: Classification with **categorical/discrete features** (non-numerical, unordered).
  - Example: Loan approval using categories like `income = {"High", "Middle", "Low"}`.
- **Formula**:

$$P(x_i|y) = \frac{\text{Count}(x_i, y) + \alpha}{\text{Count}(y) + \alpha k}$$

  - $\text{Count}(x_i, y)$: Number of times category x_i appears in class y.
  - $\text{Count}(y)$: Total samples in class y.
  - k: Number of categories for feature x_i (e.g., $k = 3$ for "High/Middle/Low").
  - α: Smoothing parameter (Laplace if $\alpha = 1$).

Naïve Bayes: Loan Approval Dataset

Age Group	Income Level	Credit Score	Loan Approved
Young	Low	Fair	No
Middle	High	Good	Yes
Senior	Medium	Good	Yes
Young	Low	Poor	No
Middle	High	Excellent	Yes
Senior	Medium	Fair	No

New Applicant: Middle-aged, High income, Good credit

Question: Approve loan (Yes/No)?

Categorical Dataset

Naïve Bayes: Loan Approval Dataset

Step 1: Calculate Priors

- Total samples = 6
  - $P(\text{Yes}) = \frac{3}{6} = 0.5$
  - $P(\text{No}) = \frac{3}{6} = 0.5$

Naïve Bayes: Loan Approval Dataset

Step 2: Compute Likelihoods

For Class "Yes" (Approved):

1. **Age Group = Middle**:
   - Count(Middle|Yes) = 2 (rows 2 & 5)
   - $P(\text{Middle}|\text{Yes}) = \frac{2}{3}$
2. **Income Level = High**:
   - Count(High|Yes) = 2 (rows 2 & 5)
   - $P(\text{High}|\text{Yes}) = \frac{2}{3}$
3. **Credit Score = Good**:
   - Count(Good|Yes) = 2 (rows 2 & 3)
   - $P(\text{Good}|\text{Yes}) = \frac{2}{3}$

Naïve Bayes: Loan Approval Dataset

Step 2: Compute Likelihoods

For Class "No" (Rejected):

1. **Age Group = Middle**:
   - Count(Middle|No) = 0 (no Middle-aged rejections)
   - $P(\text{Middle}|\text{No}) = \frac{0}{3} = 0$
2. **Income Level = High**:
   - Count(High|No) = 0
   - $P(\text{High}|\text{No}) = \frac{0}{3} = 0$
3. **Credit Score = Good**:
   - Count(Good|No) = 0
   - $P(\text{Good}|\text{No}) = \frac{0}{3} = 0$

Naïve Bayes: Loan Approval Dataset

Step 3: Posterior Probability Calculation

For class "Yes":

$$P(Yes|X) \propto P(Middle|Yes) \times P(High|Yes) \times P(Good|Yes) \times P(Yes)$$

$$= \frac{2}{3} \times \frac{2}{3} \times \frac{2}{3} \times 0.5 = \frac{8}{27} \times 0.5 \approx 0.148$$

For class "No":

$$P(No|X) \propto P(Middle|No) \times P(High|No) \times P(Good|No) \times P(No)$$

$$= 0 \times 0 \times 0 \times 0.5 = 0$$

Naïve Bayes: Loan Approval Dataset

Total evidence:

$$P(X) = P(X|Yes)P(Yes) + P(X|No)P(No) \approx 0.148 + 0 = 0.148$$

Normalized probabilities:

Step 4:

$$P(Yes|X) = \frac{0.148}{0.148} = 1.0$$

$$P(No|X) = \frac{0}{0.148} = 0$$

Prediction: "Yes" with 100% probability, since P(No|X) = 0. This shows the model becomes overconfident when encountering zero probabilities, highlighting why Laplace smoothing is typically needed in practice.

Categorical Naïve Bayes

Categorical Naïve Bayes: Loan Approval Dataset

Step 1: Calculate Priors

- Total samples = 6
  - $P(\text{Yes}) = \frac{3}{6} = 0.5$
  - $P(\text{No}) = \frac{3}{6} = 0.5$

Categorical Naïve Bayes: Loan Approval Dataset

Step 2: Compute Likelihoods with Laplace Smoothing

Laplace smoothing adds `1` to each count and `k` to the denominator, where `k` is the number of categories for each feature:

$$P(x_i|y) = \frac{\text{Count}(x_i, y) + \alpha}{\text{Count}(y) + \alpha k}$$

For Class "Yes" (Approved):

1. **Age Group = Middle**:
   - Categories: Young, Middle, Senior → $k = 3$
   - $P(\text{Middle}|\text{Yes}) = \frac{2+1}{3+3} = 0.5$
2. **Income Level = High**:
   - Categories: Low, Medium, High → $k = 3$
   - $P(\text{High}|\text{Yes}) = \frac{2+1}{3+3} = 0.5$
3. **Credit Score = Good**:
   - Categories: Fair, Good, Excellent, Poor → $k = 4$
   - $P(\text{Good}|\text{Yes}) = \frac{2+1}{3+4} \approx 0.428$

Categorical Naïve Bayes: Loan Approval Dataset

Step 2: Compute Likelihoods with Laplace Smoothing

For Class "No" (Rejected):

1. **Age Group = Middle:**
   - $P(\text{Middle}|\text{No}) = \frac{0+1}{3+3} \approx 0.166$
2. **Income Level = High:**
   - $P(\text{High}|\text{No}) = \frac{0+1}{3+3} \approx 0.166$
3. **Credit Score = Good:**
   - $P(\text{Good}|\text{No}) = \frac{0+1}{3+4} \approx 0.142$

Categorical Naïve Bayes: Loan Approval Dataset

Step 3: Calculate Posterior Probabilities

$$P(\text{Yes}|X) \propto 0.5 \times 0.5 \times 0.428 \times 0.5 \approx 0.0535$$

$$P(\text{No}|X) \propto 0.166 \times 0.166 \times 0.142 \times 0.5 \approx 0.00196$$

Normalized Posteriors:

$$P(\text{Yes}|X) = \frac{0.0535}{0.0535 + 0.00196} \approx 0.965 \quad (96.5\%)$$

$$P(\text{No}|X) = \frac{0.00196}{0.0535 + 0.00196} \approx 0.035 \quad (3.5\%)$$

Categorical Naïve Bayes: Loan Approval Dataset

Step 4: Prediction

- **Prediction**: "Yes" (Approved) with **96.5% confidence** (vs. 100% without smoothing).
- **Key Improvement**: Even though "Middle+High+Good" never appeared in "No" class, Laplace smoothing avoids zero probabilities, yielding more realistic confidence scores.

Multinomial Naïve Bayes

Multinomial Naïve Bayes: Email Spam Detection Dataset

Email Text	Contains "FREE"	Contains "Win"	Contains "Urgent"	Spam (Class)
"FREE lottery"	1	0	0	1 (Spam)
"Meeting tomorrow"	0	0	0	0 (Ham)
"Win a prize!"	0	1	0	1
"FREE urgent offer"	1	0	1	1
"Project update"	0	0	0	0
"Urgent: Your account"	0	0	1	1
"Team lunch"	0	0	0	0
"FREE trial"	1	0	0	1

Goal: Classify the email **"FREE urgent win"** (Features: FREE=1, Win=1, Urgent=1).

Multinomial Naïve Bayes: Email Spam Detection Dataset

Step 1: Calculate Priors

- Total emails: 8
  - $P(\text{Spam}) = \frac{5}{8} = 0.625$
  - $P(\text{Ham}) = \frac{3}{8} = 0.375$

Multinomial Naïve Bayes: Email Spam Detection Dataset

Step 2: Compute Likelihoods with Laplace Smoothing

Vocabulary size (k): 3 ("FREE", "Win", "Urgent")

Laplace smoothing (α=1):

$$P(x_i|Y) = \frac{\text{Count}(x_i, Y) + \alpha}{\text{Total counts in } Y + \alpha k}$$

For Spam (Y=1):

- Total word counts in Spam: 3 (FREE) + 1 (Win) + 2 (Urgent) = **6**

$$P(\text{FREE} = 1|\text{Spam}) = \frac{3+1}{6+1\times 3} = \frac{4}{9} \approx 0.444$$
$$P(\text{Win} = 1|\text{Spam}) = \frac{1+1}{6+3} = \frac{2}{9} \approx 0.222$$
$$P(\text{Urgent} = 1|\text{Spam}) = \frac{2+1}{6+3} = \frac{3}{9} \approx 0.333$$

Multinomial Naïve Bayes: Email Spam Detection Dataset

Step 2: Compute Likelihoods with Laplace Smoothing

For Ham (Y=0):

- Total word counts in Ham: 0

$$P(\text{FREE} = 1|\text{Ham}) = \frac{0+1}{0+3} = \frac{1}{3} \approx 0.333$$
$$P(\text{Win} = 1|\text{Ham}) = \frac{0+1}{0+3} = \frac{1}{3} \approx 0.333$$
$$P(\text{Urgent} = 1|\text{Ham}) = \frac{0+1}{0+3} = \frac{1}{3} \approx 0.333$$

Critical Note: Smoothing prevents zero probabilities for unseen words in Ham.

Multinomial Naïve Bayes: Email Spam Detection Dataset

Step 3: Calculate Posterior Probabilities

For email **"FREE urgent win"** (FREE=1, Win=1, Urgent=1):

Spam Probability:

$$P(\text{Spam}|X) \propto 0.625 \times 0.444 \times 0.222 \times 0.333 \approx 0.0205$$

Ham Probability:

$$P(\text{Ham}|X) \propto 0.375 \times 0.333 \times 0.333 \times 0.333 \approx 0.0139$$

Normalized Posteriors:

$$P(\text{Spam}|X) = \frac{0.0205}{0.0205 + 0.0139} \approx 0.596 \quad (59.6\%)$$
$$P(\text{Ham}|X) = \frac{0.0139}{0.0205 + 0.0139} \approx 0.404 \quad (40.4\%)$$

Multinomial Naïve Bayes: Email Spam Detection Dataset

Step 4: Prediction

- **Prediction: Spam** (59.6% confidence).

Gaussian Naïve Bayes

Gaussian Naïve Bayes: Diabetes Prediction Dataset

Glucose	BloodPressure	BMI	Age	Diabetes
148	72	33.6	50	1
85	66	26.6	31	0
183	64	23.3	32	1
89	66	23.0	21	0
137	40	43.1	33	1
116	74	25.6	30	0
78	50	31.0	26	1
115	0	35.3	29	0
197	70	45.0	53	1
125	96	0	54	1

Continuous Dataset

New Patient: Glucose=130, BloodPressure=70, BMI=32, Age=45

Goal: Predict diabetes probability.

Gaussian Naïve Bayes: Diabetes Prediction Dataset

Step 1: Calculate Priors

$$P(\text{Diabetic}) = \frac{\text{Number of diabetic patients}}{\text{Total patients}} = \frac{6}{10} = 0.6$$

$$P(\text{Non-diabetic}) = \frac{4}{10} = 0.4$$

Gaussian Naïve Bayes: Diabetes Prediction Dataset

Step 2: Calculate Mean (μ) and Standard Deviation (σ)

For Diabetic Patients (Class=1):

1. **Glucose**:
   - Values: [148, 183, 137, 78, 197, 125]
   - $\mu = \frac{148+183+137+78+197+125}{6} = 148.83$
   - $\sigma = \sqrt{\frac{(148-148.83)^2+(183-148.83)^2+\ldots}{6}} \approx 46.51$
2. **BloodPressure**:
   - Values: [72, 64, 40, 50, 70, 96]
   - $\mu = \frac{72+64+40+50+70+96}{6} = 62.33$
   - $\sigma = \sqrt{\frac{(72-62.33)^2+\ldots}{6}} \approx 24.89$

Gaussian Naïve Bayes: Diabetes Prediction Dataset

Step 2: Calculate Mean (μ) and Standard Deviation (σ)

For Diabetic Patients (Class=1):

3. **BMI**:
    - Values: [33.6, 23.3, 43.1, 31.0, 45.0, 0] (Note: 0 is likely missing data)
    - Filtered: [33.6, 23.3, 43.1, 31.0, 45.0] (remove 0)
    - $\mu = \frac{33.6+23.3+43.1+31.0+45.0}{5} = 35.0$
    - $\sigma = \sqrt{\frac{(33.6-35.0)^2+\dots}{5}} \approx 8.37$
4. **Age**:
    - Values: [50, 32, 33, 26, 53, 54]
    - $\mu = \frac{50+32+33+26+53+54}{6} = 39.67$
    - $\sigma = \sqrt{\frac{(50-39.67)^2+\dots}{6}} \approx 12.16$

Gaussian Naïve Bayes: Diabetes Prediction Dataset

Step 2: Calculate Mean (μ) and Standard Deviation (σ)

For Non-diabetic Patients (Class=0):

1. **Glucose**:
   - Values: [85, 89, 116, 115]
   - $\mu = \frac{85+89+116+115}{4} = 101.25$
   - $\sigma = \sqrt{\frac{(85-101.25)^2+\dots}{4}} \approx 15.36$
2. **BloodPressure**:
   - Values: [66, 66, 74, 0] (Note: 0 is likely missing data)
   - Filtered: [66, 66, 74]
   - $\mu = \frac{66+66+74}{3} \approx 68.67$
   - $\sigma = \sqrt{\frac{(66-68.67)^2+\dots}{3}} \approx 4.62$

Gaussian Naïve Bayes: Diabetes Prediction Dataset

Step 2: Calculate Mean (μ) and Standard Deviation (σ)

For Non-diabetic Patients (Class=0):

3. **BMI**:
   - Values: [33.6, 23.3, 43.1, 31.0, 45.0, 0] (Note: 0 is likely missing data)
   - Filtered: [33.6, 23.3, 43.1, 31.0, 45.0] (remove 0)
   - $\mu = \frac{33.6+23.3+43.1+31.0+45.0}{5} = 35.0$
   - $\sigma = \sqrt{\frac{(33.6-35.0)^2+\ldots}{5}} \approx 8.37$
4. **Age**:
   - Values: [50, 32, 33, 26, 53, 54]
   - $\mu = \frac{50+32+33+26+53+54}{6} = 39.67$
   - $\sigma = \sqrt{\frac{(50-39.67)^2+\ldots}{6}} \approx 12.16$

Gaussian Naïve Bayes: Diabetes Prediction Dataset

Step 3: Calculate Likelihoods Using Gaussian PDF

$$P(x_i|Y) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} \exp\left(-\frac{(x_i - \mu_Y)^2}{2\sigma_Y^2}\right)$$

Gaussian Probability Density Function (PDF)

It describes the likelihood of a continuous random variable

For Diabetic Class (1):

1. **Glucose (130)**:

$$P(130|1) = \frac{1}{\sqrt{2\pi \cdot 46.51^2}} \exp\left(-\frac{(130 - 148.83)^2}{2 \cdot 46.51^2}\right) \approx 0.0085$$

2. **BloodPressure (70)**:

$$P(70|1) = \frac{1}{\sqrt{2\pi \cdot 24.89^2}} \exp\left(-\frac{(70 - 62.33)^2}{2 \cdot 24.89^2}\right) \approx 0.015$$

3. **BMI (32)**:

$$P(32|1) = \frac{1}{\sqrt{2\pi \cdot 8.37^2}} \exp\left(-\frac{(32 - 30.0)^2}{2 \cdot 8.37^2}\right) \approx 0.046$$

4. **Age (45)**:

$$P(45|1) = \frac{1}{\sqrt{2\pi \cdot 12.16^2}} \exp\left(-\frac{(45 - 39.17)^2}{2 \cdot 12.16^2}\right) \approx 0.028$$

5. **Combined Likelihood**:

$$P(X|1) = 0.0085 \times 0.015 \times 0.046 \times 0.028 \approx 1.64 \times 10^{-7}$$

Gaussian Naïve Bayes: Diabetes Prediction Dataset

Step 3: Calculate Likelihoods Using Gaussian PDF

$$P(x_i|Y) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} \exp\left(-\frac{(x_i - \mu_Y)^2}{2\sigma_Y^2}\right)$$

For Non-diabetic Class (0):

1. **Glucose (130)**:

$$P(130|0) = \frac{1}{\sqrt{2\pi \cdot 15.36^2}} \exp\left(-\frac{(130 - 101.25)^2}{2 \cdot 15.36^2}\right) \approx 0.00002$$

2. **BloodPressure (70)**:

$$P(70|0) = \frac{1}{\sqrt{2\pi \cdot 31.92^2}} \exp\left(-\frac{(70 - 51.5)^2}{2 \cdot 31.92^2}\right) \approx 0.012$$

3. **BMI (32)**:

$$P(32|0) = \frac{1}{\sqrt{2\pi \cdot 9.19^2}} \exp\left(-\frac{(32 - 21.58)^2}{2 \cdot 9.19^2}\right) \approx 0.0007$$

4. **Age (45)**:

$$P(45|0) = \frac{1}{\sqrt{2\pi \cdot 3.59^2}} \exp\left(-\frac{(45 - 27.75)^2}{2 \cdot 3.59^2}\right) \approx 0$$

5. **Combined Likelihood**:

$$P(X|0) \approx 0 \quad \text{(Age dominates)}$$

Gaussian Naïve Bayes: Diabetes Prediction Dataset

Step 4: Posterior Probabilities

$$P(1|X) \propto P(X|1) \times P(1) = 1.64 \times 10^{-7} \times 0.6 \approx 9.84 \times 10^{-8}$$

$$P(0|X) \propto P(X|0) \times P(0) = 0 \times 0.4 = 0$$

Normalized:

$$P(1|X) = \frac{9.84 \times 10^{-8}}{9.84 \times 10^{-8} + 0} = 1.0 \quad (100\%)$$

$$P(0|X) = 0$$

Prediction: **Diabetic** (due to high Glucose/BMI and Age mismatch with non-diabetic class).

Zero probability for Age in non-diabetic class leads to overconfidence.

Solution: Use kernel density estimation or Laplace smoothing (for categorical features).

When to Use/Avoid Naïve Bayes?

When Use:

1. **Text Data** (spam, reviews) → Fast & works well
2. **Quick Testing** → Trains in seconds
3. **Categories** (like survey answers) → Handles yes/no choices easily
4. **Many Features** (e.g., 10,000+ words) → Stays efficient

Avoid When:

1. **Features Depend on Each Other** (e.g., "age + income" together matter)
2. **Complex Numbers** (like skewed income data) → Needs normal distributions
3. **Tiny Datasets** → Can break without smoothing (`alpha=1` fixes this)

Any Question?